ASSIGNMENT 5.1 & 6

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Task 1:
Employee Data: Create Python code that defines a class named `Employee` with the following attributes: `empid`, `empname`, `designation`, `basic_salary`, and `exp`. Implement a method `display_details()` to print all employee details. Implement another method `calculate_allowance()` to determine additional allowance based on experience:
- If `exp > 10 years` → allowance = 20% of `basic_salary`
- If `5 ≤ exp ≤ 10 years` → allowance = 10% of `basic_salary`
- If `exp < 5 years` → allowance = 5% of `basic_salary`

Finally, create at least one instance of the `Employee` class, call the `display_details()` method, and print the calculated allowance.

```
PS C:\Users\ADMIN> & "C:/Program Files/Pyt
Employee ID: 101
Employee Name: Alice
Designation: Manager
Basic Salary: 80000
Experience: 12 years
Allowance: 16000.0
Total Salary: 96000.0
```

Task 2:

Electricity Bill Calculation- Create Python code that defines a class named `ElectricityBill` with attributes: `customer_id`, `name`, and `units_consumed`. Implement a method `display_details()` to print customer details, and a method `calculate_bill()` where:

- Units ≤ 100 → ₹5 per unit
- 101 to 300 units → ₹7 per unit
- More than 300 units → ₹10 per unit

Create a bill object, display details, and print the total bill amount.

```
> Users > ADMIN > OneDrive > Desktop > certificates > LAB 5.1.py > ElectricityBill > calculate_bill
33  class ElectricityBill:
34      def __init__(self,customer_id,name,units_consumed):
35          self.customer_id = customer_id
36          self.name = name
37          self.units_consumed = units_consumed
38      def display_bill(self):
39          print(f"Customer ID: {self.customer_id}")
40          print(f"Customer Name: {self.name}")
41          print(f"Units Consumed: {self.units_consumed}")
42      def calculate_bill(self):
43          if self.units_consumed <= 100:
44              rate = 5
45          elif self.units_consumed <=300:
46              rate = 7
47          else:
48              rate = 10
49          print(f"Rate per unit: {rate}")
50          print(f"Total Bill Amount: {self.units_consumed * rate}")
51  billobj1=ElectricityBill(201,"David",80)
52  billobj1.display_bill()
53  billobj1.calculate_bill()
```

```
Customer ID: 201
Customer Name: David
Units Consumed: 80
Rate per unit: 5
Total Bill Amount: 400
PS C:\Users\ADMIN>
```

Task 3:

Product Discount Calculation- Create Python code that defines a class named `Product` with attributes: `product_id`, `product_name`, `price`, and `category`. Implement a method `display_details()` to print product details. Implement another method `calculate_discount()` where:

- Electronics → 10% discount

- Clothing → 15% discount
- Grocery → 5% discount

Create at least one product object, display details, and print the final price after discount.

```
class Product:
    def __init__(self,product_id,product_name,price,category):
        self.product_id = product_id
        self.product_name = product_name
        self.price = price
        self.category = category
    def display_details(self):
        print(f"Product ID: {self.product_id}")
        print(f"Product Name: {self.product_name}")
        print(f"Price: {self.price}")
        print(f"Category: {self.category}")
    def calculate_discount(self):
        if self.category.lower() == "electronics":
            discount = 0.10 * self.price
        elif self.category.lower() == "clothing":
            discount = 0.15 * self.price
        else:
            discount = 0.05 * self.price
        print(f"Discount: {discount}")
        print(f"Price after Discount: {self.price - discount}")
productobj1 = Product(301,"Smartphone",50000,"Electronics")
productobj1.display_details()
final_price = productobj1.calculate_discount()
print(final_price)
```

```
Product ID: 301
Product Name: Smartphone
Price: 50000
Category: Electronics
Discount: 5000.0
Price after Discount: 45000.0
None
```

Task 4:

Book Late Fee Calculation- Create Python code that defines a class named `LibraryBook` with attributes: `book_id`, `title`, `author`, `borrower`, and `days_late`. Implement a method `display_details()` to print book details, and a method `calculate_late_fee()` where:

- Days late ≤ 5 → ₹5 per day
- 6 to 10 days late → ₹7 per day
- More than 10 days late → ₹10 per day

Create a book object, display details, and print the late fee.

```python
class LibraryBook:
    def __init__(self,book_id,title,author,borrower,days_late):
        self.book_id = book_id
        self.title = title
        self.author = author
        self.borrower = borrower
        self.days_late = days_late
    def display_details(self):
        print(f"Book ID: {self.book_id}")
        print(f"Title: {self.title}")
        print(f"Author: {self.author}")
        print(f"Borrower: {self.borrower}")
        print(f"Days Late: {self.days_late}")
    def calculate_late_fee(self):
        if self.days_late <=5 :
            late_fee = self.days_late * 5
        elif self.days_late <=10:
            late_fee = self.days_late * 7
        else:
            late_fee = self.days_late * 10
        print(f"Late Fee: {late_fee}")
bookobj1 = LibraryBook(401,"1984","George Orwell","Eve",8)
bookobj1.display_details()
print(bookobj1.calculate_late_fee())
```

```
Book ID: 401
Title: 1984
Author: George Orwell
Borrower: Eve
Days Late: 8
Late Fee: 56
```

Task 5:

Student Performance Report - Define a function `student_report(student_data)` that accepts a dictionary containing student names and their marks. The function should:

- Calculate the average score for each student
- Determine pass/fail status (pass ≥ 40)
- Return a summary report as a list of dictionaries

Use Copilot suggestions as you build the function and format the output.

```python
class StudentReport:
    def __init__(self, name, marks):
        self.name = name
        self.marks = marks

    def average_grade(self):
        return sum(self.marks) / len(self.marks)

    def report(self):
        avg = self.average_grade()
        return f"Student: {self.name}, Average Grade: {avg:.2f}"
    def determine_pass_fail(self):
        avg = self.average_grade()
        return "Pass" if avg >= 40 else "Fail"
def report_card(student):
    print(student.report())
    print("Result:", student.determine_pass_fail())
```

```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  POR

Number of Students Passed: 8
Number of Students Failed: 2
Highest Score: 91
Lowest Score: 33
Average Score: 63.00
PS C:\Users\ADMIN>
```

Task 6:

Taxi Fare Calculation-Create Python code that defines a class named `TaxiRide` with attributes: `ride_id`, `driver_name`, `distance_km`, and `waiting_time_min`. Implement a method `display_details()` to print ride details, and a method `calculate_fare()` where:

- ₹15 per km for the first 10 km
- ₹12 per km for the next 20 km
- ₹10 per km above 30 km
- Waiting charge: ₹2 per minute

Create a ride object, display details, and print the total fare.

Users > ADMIN > OneDrive > Desktop > certificates > LAB 5.1.py > ...

```python
class TaxiRide :
    def __init__(self, ride_id, driver_name, distance_km,waiting_time_min):
        self.ride_id = ride_id
        self.driver_name = driver_name
        self.distance_km = distance_km
        self.waiting_time_min = waiting_time_min
    def display_details(self):
        print(f"Ride ID: {self.ride_id}")
        print(f"Driver Name: {self.driver_name}")
        print(f"Distance (km): {self.distance_km}")
        print(f"Waiting Time (min): {self.waiting_time_min}")
        print(f"Ride details: {self.ride_id}, {self.driver_name}, {self.distance_km} km, {self.waiting_time_min} mi
    def calculate_fare(self):
        fare = 0
        if self.distance_km <= 10:
            fare += self.distance_km * 15
        elif self.distance_km <= 30:
            fare += 10 * 15 + (self.distance_km - 10) * 12
        else:
            fare += 10 * 15 + 20 * 12 + (self.distance_km - 30) * 10
        fare += self.waiting_time_min * 2
        return fare
# Example usage:
ride = TaxiRide("R123", "John Doe", 25, 5)
ride.display_details()
fare = ride.calculate_fare()
print(f"Total Fare: ${fare:.2f}")
```

```
None
Ride ID: R123
Driver Name: John Doe
Distance (km): 25
Waiting Time (min): 5
Ride details: R123, John Doe, 25 km, 5 min
Total Fare: $340.00
PS C:\Users\ADMIN>
```

Task 7:

Statistics Subject Performance - Create a Python function `statistics_subject(scores_list)` that accepts a list of 60 student scores and computes key performance statistics. The function should return the following:

- Highest score in the class
- Lowest score in the class
- Class average score
- Number of students passed (score ≥ 40)
- Number of students failed (score < 40)

Allow Copilot to assist with aggregations and logic

```
: > Users > ADMIN > OneDrive > Desktop > certificates > LAB 5.1.py > ...
132
133   class StatisticsSubjectPerformance:
134       def statistics_subject(self, scores_list):
135           if not scores_list:
136               print("No scores available.")
137               return
138
139           highest_score = max(scores_list)
140           lowest_score = min(scores_list)
141           average_score = sum(scores_list) / len(scores_list)
142           passed_count = sum(1 for score in scores_list if score >= 40)
143           failed_count = len(scores_list) - passed_count
144           print(f"Number of Students Passed: {passed_count}")
145           print(f"Number of Students Failed: {failed_count}")
146           print(f"Highest Score: {highest_score}")
147           print(f"Lowest Score: {lowest_score}")
148           print(f"Average Score: {average_score:.2f}")
149   # Example usage
150   scores = [85, 42, 78, 90, 33, 67, 49, 58, 91, 37]
151   performance = StatisticsSubjectPerformance()
152   performance.statistics_subject(scores)
```

```
Number of Students Passed: 8
Number of Students Failed: 2
Highest Score: 91
Lowest Score: 33
Average Score: 63.00
PS C:\Users\ADMIN>
```

Lab 5: Ethical Foundations – Responsible AI Coding Practices

Lab Objectives:

• To explore the ethical risks associated with AI-generated code.
• To recognize issues related to security, bias, transparency, and copyright.
• To reflect on the responsibilities of developers when using AI tools in software development.
• To promote awareness of best practices for responsible and ethical AI coding.

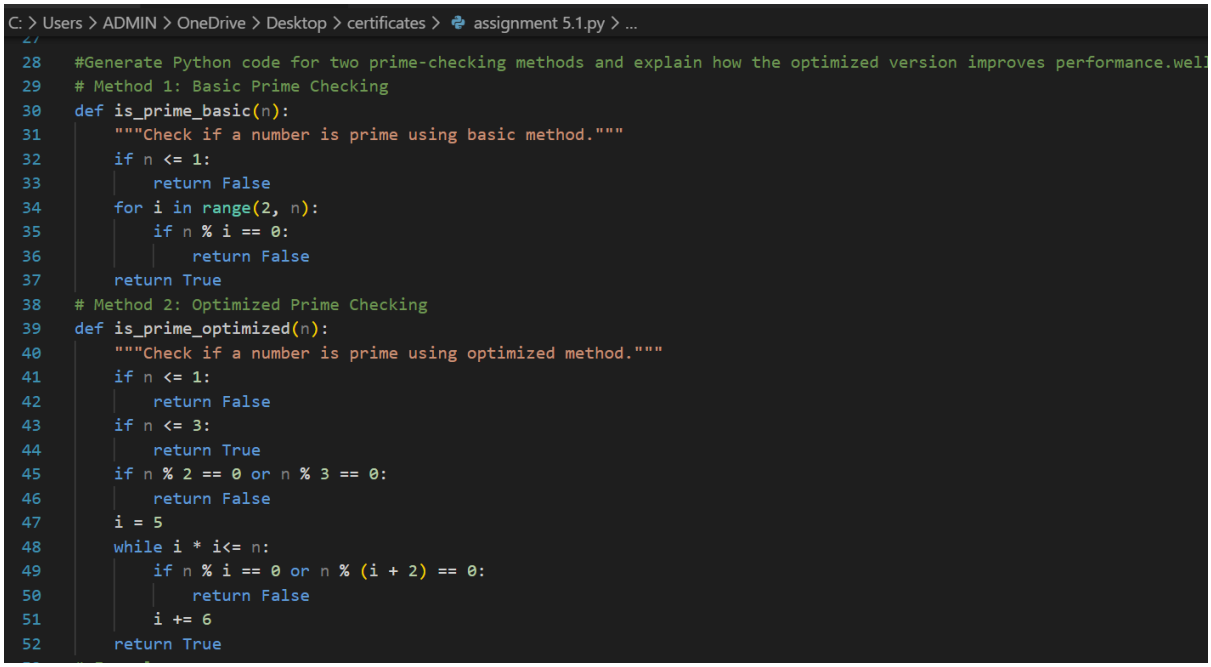
Lab Outcomes (LOs):

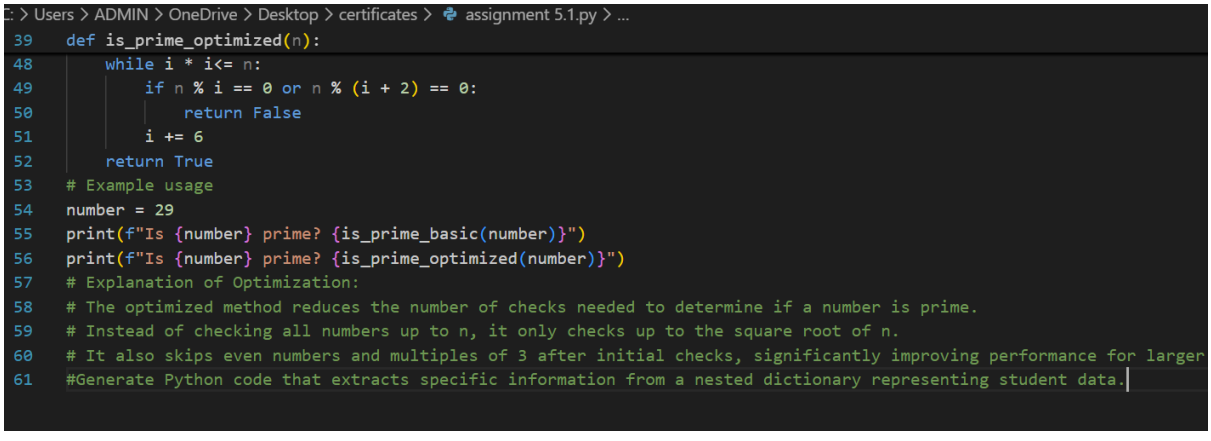
After completing this lab, students will be able to:

• Identify and avoid insecure coding patterns generated by AI tools.
• Detect and analyze potential bias or discriminatory logic in AI-generated outputs.
• Evaluate originality and licensing concerns in reused AI-generated code.
• Understand the importance of explainability and transparency in AI-assisted programming.

• Reflect on accountability and the human role in ethical AI coding practices.

Task Description #8 (Transparency in Algorithm Optimization)
Task: Use AI to generate two solutions for checking prime numbers:
• Naive approach(basic)
• Optimized approach
Prompt:
“Generate Python code for two prime-checking methods and explain how the optimized version improves performance.”
Expected Output:
• Code for both methods.
• Transparent explanation of time complexity.
• Comparison highlighting efficiency improvements.





Task Description #9 (Transparency in Recursive Algorithms)
Objective: Use AI to generate a recursive function to calculate Fibonacci numbers.
Instructions:
1. Ask AI to add clear comments explaining recursion.
2. Ask AI to explain base cases and recursive calls.
Expected Output:
• Well-commented recursive code.
• Clear explanation of how recursion works.
• Verification that explanation matches actual execution.

C: > Users > ADMIN > OneDrive > Desktop > certificates > assignment 5.1.py > fibonacci

```python
#generate a fibonacci series using recursion , well decummented and well commented code
def fibonacci(n):
    """
    This function returns the nth Fibonacci number using recursion.

    Parameters:
    n (int): The position in the Fibonacci series to retrieve. Must be a non-negative integer.

    Returns:
    int: The nth Fibonacci number.

    Raises:
    ValueError: If n is a negative integer.
    """
    # Check if the input is a negative integer
    if n < 0:
        raise ValueError("Input must be a non-negative integer.")
    # Base case: the first Fibonacci number is 0
    if n == 0:
        return 0
    # Base case: the second Fibonacci number is 1
    elif n == 1:
        return 1
    # Recursive case: the nth Fibonacci number is the sum of the (n-1)th and (n-2)th Fibonacci numbers
    else:
        return fibonacci(n - 1) + fibonacci(n - 2)

```

Task Description #10 (Transparency in Error Handling)
Task: Use AI to generate a Python program that reads a file and processes data.

Prompt:
“Generate code with proper error handling and clear explanations for each exception.”
Expected Output:
• Code with meaningful exception handling.
• Clear comments explaining each error scenario.
• Validation that explanations align with runtime behavior.

```python
#generate a Python program that reads a file and processes data with proper error handling and clear explanations for
def read_and_process_file(file_path):
    try:
        # Attempt to open the file
        with open(file_path, 'r') as file:
            data = file.readlines()

        # Process the data (for example, convert each line to an integer and sum them)
        total = 0
        for line in data:
            try:
                number = int(line.strip())
                total += number
            except ValueError as ve:
                print(f"ValueError: Could not convert line to integer: '{line.strip()}'. Skipping this line.")

        print(f"The total sum of the numbers in the file is: {total}")

    except FileNotFoundError as fnfe:
        print(f"FileNotFoundError: The file at path '{file_path}' was not found. Please check the path and try again.
    except IOError as ioe:
        print(f"IOError: An error occurred while trying to read the file: {ioe}")
    except Exception as e:
        print(f"An unexpected error occurred: {e}")
# Example usage
file_path = 'data.txt'  # Replace with your file path
read_and_process_file(file_path)
```